

Performance Testing

Date	15 March 2026
Team ID	xxxxxx
Project Name	Human Development index Predictor
Maximum Marks	5 Marks

Step 1: Testing Overview

Testing Tool Used: Apache JMeter, Postman

Type of Testing: Load Testing, Stress Testing

Target Module/API: HDI Prediction API, Country Data Processing Module, Classification Module

Test Environment: Flask Backend, Machine Learning Model, SQLite/MySQL Database

Test Date: 14 July 2026

Step 2: Test Scenarios

S.No	Test Scenario	Virtual Users	Duration (sec)	Expected Outcome
1	User Login	50	60	Successful login within 2 seconds
2	Country Data Submission	100	120	Data processed without errors
3	HDI Prediction	75	90	Accurate HDI prediction generated
4	HDI Classification	50	60	Correct classification returned

Step 3: Performance Test Results

Metric	Target Value	Actual Value	Status	Remarks
Average Response Time	< 2 sec	1.2 sec	Pass	Meets requirement
Maximum Response Time	< 5 sec	3.1 sec	Pass	Stable under load
Throughput	>100 req/sec	140 req/sec	Pass	Good throughput
Error Rate	<1%	0.1%	Pass	No significant failures
CPU Utilization	<80%	62%	Pass	Efficient usage
Memory Utilization	<80%	68%	Pass	Within limits

Step 4: Observations & Analysis

- The HDI Predictor maintained stable performance under normal and peak loads.
- Prediction requests were processed efficiently with low latency.
- No critical errors or server crashes were observed.
- Resource utilization remained within acceptable thresholds.
- The application is suitable for moderate concurrent users.